The Exact Size of the Conflict-Free Gadget Hierarchy

A closed form and matching growth rate for $|G_k|$

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2026-09-05

Abel et al. (2018) construct a family of graphs G_k with $\chi_{CF}(G_k) = k$, used as the forcing gadget in their proof that conflict-free coloring is NP-complete for every fixed $k \geq 3$. They bound the size of this family between $\Omega(2^k)$ and $\mathcal{O}(2^{k \log k})$. We give a closed form for $|G_k|$ by reducing the size recurrence to a Pell recurrence, and show $|G_k| = \Theta\Big(k!\,(1+\sqrt{2})^k\Big)$, with the normalized ratio $|G_k|/(k!\,\alpha^k)$ converging to an explicit constant. The rate is consistent with both published bounds and strictly sharper than either.

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1 Setup

Sections Section 2 through Section 8 are reproduced verbatim from the longer walkthrough. This section is the only new material, and exists to make the note self-contained.

We recall the two definitions the result depends on, and nothing else.

Definition 1.1 (Conflict-free coloring)**.** A **partial k-coloring** of G is an assignment $\chi : V' \to \{1, \dots, k\}$ for some subset $V' \subseteq V(G)$; vertices outside V' are **uncolored**. Such a χ is **conflict-free** with respect to closed neighborhoods if for every $v \in V(G)$ there is a vertex $w \in N[v] \cap V'$ whose color is unique in $N[v]$ — that is, $\chi(w') \neq \chi(w)$ for every other $w' \in N[v] \cap V'$. We call w the **conflict-free neighbor** of v. The **conflict-free chromatic number** $\chi_{CF}(G)$ is the smallest k for which such a coloring exists.

The family G_k is built recursively, with G_1 a single vertex and G_2 the star $K_{1,3}$ with one edge subdivided. Given G_k and G_{k-1}:

- Take a complete graph $G = K_{k+1}$ on $k+1$ vertices.
- To each vertex $v \in V(K_{k+1})$ attach two disjoint and independent copies of G_k, adding an edge from every v to every vertex of both copies of G_k.
- For each edge $e = vw \in E(K_{k+1})$, add two disjoint and independent copies of G_{k-1}, where there is an edge from every v and w to every vertex on both copies of G_{k-1}.

This gives $\chi_{CF}(G_k) = k$ for all $k \geq 1$. Only the vertex counts matter below.

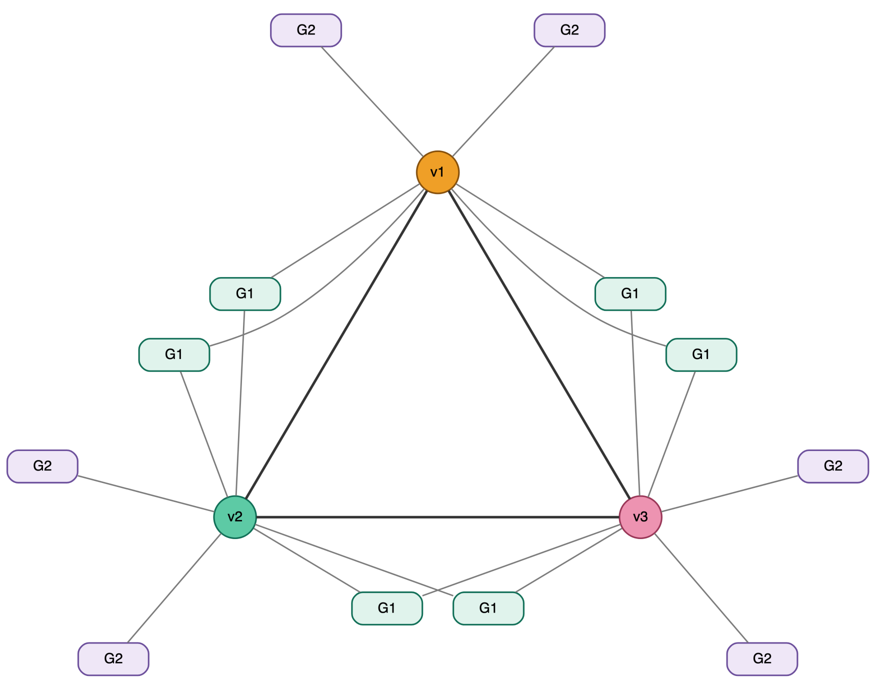

Figure 1: G_3: a K_3 core, two copies of G_2 on each vertex, two copies of G_1 on each edge.

2 Counting the vertices

$$|G_1| = 1, \qquad |G_2| = 5, \qquad |G_3| = 3 + 3(2|G_2|) + 3(2|G_1|) = 3 + 30 + 6 = 39.$$

In general, the core contributes $k+1$ vertices, each of the $k+1$ core vertices carries $2|G_k|$, and each of the $\binom{k+1}{2}$ core edges carries $2|G_{k-1}|$. Since

$$\binom{k+1}{2} = \frac{(k+1)!}{2!\,(k-1)!} = \frac{k(k+1)}{2},$$

we get

$$|G_{k+1}| \;=\; \underbrace{(k+1)}_{v \in K_{k+1}} \;+\; \underbrace{2(k+1)\,|G_k|}_{H \text{ graphs}} \;+\; \underbrace{k(k+1)\,|G_{k-1}|}_{J \text{ graphs}}$$

and factoring out $(k+1)$,

$$\boxed{|G_{k+1}| = (k+1)\Big(2|G_k| + k\,|G_{k-1}| + 1\Big).}$$

Check: $|G_3| = 3(2 \cdot 5 + 2 \cdot 1 + 1) = 3 \cdot 13 = 39$ and $|G_4| = 4(2 \cdot 39 + 3 \cdot 5 + 1) = 4 \cdot 94 = 376$.

3 Algorithmic bounds

Every term of the bracket is positive, so $a_{k+1} \geq 2a_k$, and therefore

$$a_k = \Omega(2^k)\,.$$

This is the bound stated in the paper, and it is all that is needed there. We can do better by keeping the factor in front: since $a_{k+1} \geq 2(k+1)a_k$, unrolling from $a_2 = 5$ gives

$$a_k \;\geq\; 5\prod_{j=3}^{k} 2j \;=\; 5 \cdot 2^{k-2} \cdot \frac{k!}{2},$$

so $\log_2 a_k = \Omega(k \log k)$.

The sequence is increasing, so $a_{k-1} \leq a_k$, and also $1 \leq a_k$. Hence

$$a_{k+1} \;\leq\; (k+1)(2a_k + k\,a_k + a_k) \;=\; (k+1)(k+3)\,a_k \;\leq\; (k+3)^2 a_k.$$

Unrolling from $a_1 = 1$,

$$a_{k+1} \;\leq\; \prod_{j=1}^{k} (j+3)^2 \;\leq\; (k+3)^{2k},$$

and taking logarithms,

$$\log_2 a_{k+1} \;\le\; 2k \log_2(k+3) \;=\; O(k \log k),$$

so $a_k = \mathcal{O}(2^{k \log k})$.

3.1 Solving the recurrence

First we rewrite our recurrence as a sequence:

$$a_1 = 1, \qquad a_2 = 5, \qquad a_{k+1} = (k+1)\big(2a_k + k\,a_{k-1} + 1\big), \qquad k, a \in \mathbb{Z}^+.$$

The $(k+1)$ out front and the k on the second term hide a Pell recurrence. We divide it out.

4 Normalizing away the factorials

Divide both sides by $(k+1)!$:

$$\frac{a_{k+1}}{(k+1)!} \;=\; \frac{2a_k + k\,a_{k-1} + 1}{k!}.$$

Let $b_k = \dfrac{a_k}{k!}$. The middle term simplifies because $\dfrac{k\,a_{k-1}}{k!} = \dfrac{a_{k-1}}{(k-1)!} = b_{k-1}$, so

$$b_{k+1} \;=\; 2b_k + b_{k-1} + \frac{1}{k!}.$$

Isolating the $1/k!$, what is left is exactly the **Pell recurrence**

$$b_{k+1} = 2b_k + b_{k-1}.$$

The seeds come from the graph: $b_1 = \dfrac{a_1}{1!} = 1$, where a_1 is the number of vertices in G_1, and $b_2 = \dfrac{a_2}{2!} = \dfrac{5}{2}$, referencing $K_{1,3}$ for a_2.

5 The Pell recurrence

$$P_n = 2P_{n-1} + P_{n-2}, \qquad P_0 = 0,\ P_1 = 1, \qquad \langle 0, 1, 2, 5, 12, 29, 70, \dots \rangle$$

Set $P_n = r^n$, so $r^n = 2r^{n-1} + r^{n-2}$, giving $r^2 = 2r + 1$ and

$$r^2 - 2r - 1 = 0, \qquad r = \frac{2 \pm \sqrt{4+4}}{2} = \frac{2 \pm \sqrt{8}}{2}, \qquad r = 1 + \sqrt{2},\ r = 1 - \sqrt{2}.$$

The recurrence is linear, so

$$P_n = A(1+\sqrt{2})^n + B(1-\sqrt{2})^n$$

where A and B satisfy all values of the recurrence. From $P_0 = 0$ we get $A = -B$, and from $P_1 = 1$,

$$1 = A(1+\sqrt{2}) - A(1-\sqrt{2}) = A(1+\sqrt{2}+\sqrt{2}-1) = A \cdot 2\sqrt{2},$$

so $A = \dfrac{1}{2\sqrt{2}}$ and $B = \dfrac{-1}{2\sqrt{2}}$, giving

$$P_n = \frac{(1+\sqrt{2})^n - (1-\sqrt{2})^n}{2\sqrt{2}}.$$

5.1 The homogeneous part

Running the Pell recurrence forward from b_1, b_2:

$$b_3 = 2b_2 + b_1, \qquad b_4 = 2b_3 + b_2 = 5b_2 + 2b_1, \qquad b_5 = 2b_4 + b_3 = 12b_2 + 5b_1.$$

The coefficients are Pell numbers, so we generalize:

$$b_k \;=\; P_{k-1}\, b_2 + P_{k-2}\, b_1 \;=\; \frac{5}{2} P_{k-1} + P_{k-2}.$$

5.2 The inhomogeneous part

Our actual recurrence has $1/k!$ added, so

$$b_3 = 6 + \frac{1}{2!} = 6.5, \qquad b_4 = 2b_3 + b_2 + \frac{1}{3!} = 5b_2 + 2b_1 + \frac{1}{3!} + 2\left(\frac{1}{2!}\right),$$

$$b_5 = 2b_4 + b_3 + \frac{1}{4!} = 12b_2 + 5b_1 + 2\left(\frac{1}{3!}\right) + 5\left(\frac{1}{2!}\right) + \frac{1}{4!}.$$

Each factorial creates its own Pell: a term $\frac{1}{j!}$ injected at step j propagates forward with Pell coefficients, contributing $\frac{P_{k-j}}{j!}$ to b_k.

Verification. Let $k = 5$, $j = 2$. We have $\frac{P_3}{2!}$, where $P_3 = 5$, so $\frac{5}{2!}$ — which is the coefficient appearing on $\frac{1}{2!}$ in b_5 above.

Accounting for every j, for some k:

$$\sum_{j=2}^{k-1} \frac{P_{k-j}}{j!}.$$

6 The closed form

Solving for our altered Pell recurrence:

$$b_k \;=\; \frac{5P_{k-1}}{2} + P_{k-2} + \sum_{j=2}^{k-1} \frac{P_{k-j}}{j!}.$$

Remember that $b_k = \frac{a_k}{k!}$, and that a_k is just our actual recurrence sequentialized, so

$$\boxed{|G_k| \;=\; k!\left[P_{k-2} + \frac{5P_{k-1}}{2} + \sum_{j=2}^{k-1} \frac{P_{k-j}}{j!}\right]}$$

with P_n as above. Using the standard backward extension $P_{-1} = 1$, this is valid for all $k \geq 1$.

Check. $k = 3$ gives $6\left[2 + \frac{5}{2} + \frac{P_1}{2!}\right] = 6(6.5) = 39$, and $k = 4$ gives $24\left[5 + \frac{25}{2} + \frac{P_2}{2!} + \frac{P_1}{3!}\right] = 24\left(\frac{47}{3}\right) = 376$, matching the direct counts. The sequence continues 1, 5, 39, 376, 4545, 65826, 1112461, ...

7 Growth rate

Write $\alpha = 1+\sqrt{2}$ and $\beta = 1-\sqrt{2}$, so

$$P_n = \frac{\alpha^n - \beta^n}{2\sqrt{2}}.$$

Since $|\beta| = \sqrt{2}-1 < 1$, the β^n term vanishes and

$$\lim_{n\to\infty} P_n = \lim_{n\to\infty} \frac{\alpha^n}{2\sqrt{2}}, \qquad \text{so} \qquad P_n = \Theta(\alpha^n) = \Theta\big((1+\sqrt{2})^n\big).$$

For the sum, $P_{k-j} = \Theta(\alpha^{k-j})$, so

$$\sum_{j=2}^{k-1} \frac{P_{k-j}}{j!} \;=\; \Theta\left(\sum_{j=2}^{k-1} \frac{\alpha^k}{j!\,\alpha^j}\right) \;=\; \Theta\left(\alpha^k \sum_{j=2}^{k-1} \frac{1}{j!\,\alpha^j}\right),$$

where $\sum_{j\geq 2} \dfrac{1}{j!\,\alpha^j}$ converges — to $e^{1/\alpha} - 1 - \frac{1}{\alpha}$ — so the whole sum is $\Theta(\alpha^k)$. (The $j=2$ term alone is already $\Theta(\alpha^k)$, which gives the matching lower bound.)

The remaining pieces $\dfrac{5P_{k-1}}{2}$ and P_{k-2} are Pell recurrences multiplied by the constants $\dfrac{5}{2\alpha}$ and $\dfrac{1}{\alpha^2}$ respectively, so they are both $\Theta(\alpha^k)$ as well.

So $\dfrac{|G_k|}{k!} = \Theta\Big((1+\sqrt{2})^k\Big)$, and thus

$$\boxed{|G_k| \;=\; \Theta\Big(k!\,(1+\sqrt{2})^k\Big).}$$

Because every piece has the same growth constant, the ratio actually converges rather than merely being bounded:

$$\lim_{k\to\infty} \frac{|G_k|}{k!\,\alpha^k} \;=\; \frac{1}{2\sqrt{2}}\left(e^{\sqrt{2}-1} + \frac{1-\sqrt{2}}{2}\right) \;\approx\; 0.4617667,$$

using $1/\alpha = \sqrt{2}-1$.

8 Comparison with the paper

The paper states only that $|G_k|$ is in $\Omega(2^k)$ and $\mathcal{O}(2^{k\log k})$, for **fixed** k the gadget has constant size, which is what makes the reduction in Lemma 3.4 run in polynomial time. The exact rate is consistent with both bounds, since $\log_2(k!\,\alpha^k) = \Theta(k\log k)$, but the closed form derived in this document along with it's growth raate are strictly sharper because we pin the base at $1+\sqrt{2} \approx 2.414$

What the sharper rate buys. The constant hidden in the reduction of (Abel et al. 2018, Lemma 3.4) is $C_1 = 2|G_k|$. Under the closed form this is $\Theta\Big(k!\,(1+\sqrt{2})^k\Big)$, already 752 at $k = 4$. The reduction is therefore linear for each fixed k and catastrophic in k, which is why the result is a statement about CF k-Coloring for each constant k rather than about the problem with k given as part of the input.

9 References

Abel, Zachary, Victor Alvarez, Erik D. Demaine, et al. 2018. "Conflict-Free Coloring of Graphs." *SIAM Journal on Discrete Mathematics* 32 (4): 2675–702. https://doi.org/10.1137/17M1146579.